WELLS FARGO

Generative AI for grounded procedure guidance.

Conversational access to eleven operational procedures, with traceable answers and contextual follow-ups.

BUSINESS OBJECTIVE

Help branch employees find and verify applicable guidance across long, conditional procedures.

SOLUTION METHODOLOGY

Retrieve supporting passages, generate a concise answer, then check each claim against its cited source.

IMPLEMENTED CAPABILITIES

Generative chat

Answers from procedure evidence with source citations.

Follow-up handling

Retain the situation and retrieve fresh evidence.

Evidence workspace

Original text, answer checks and execution activity.

Quality evaluation

Retrieval comparisons and response-quality diagnostics.

11 source procedures **139** HTML tables **1,388** indexed passages

Value to measure: accurate, relevant, complete and concise guidance with less search effort.

Implementation scope: provided Cayman Bank procedures · local model inference · source citations and reviewable check records

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From original documents to structured, searchable evidence.

Preserve content, conditions and source identity throughout ingestion, chunking and indexing.

01 / PROFILE

Inventory and profile

11 HTML files; 79,424 extracted words; 139 tables. Capture titles, IDs and publication details.

02 / EXTRACT

Parse structure

UTF-8 headings, lists and table rows. Preserve column labels, row conditions and nested content.

03 / ORGANIZE

Chunk with context

~110 narrative words; 20-word overlap. Repeat table conditions on continuation passages.

04 / INDEX

Index and trace

1,388 passages; BM25 + 384D dense vectors. Record source, section and original-file hash.

DESIGN RATIONALE

Focused passages improve retrieval; repeated conditions protect applicability. Source fingerprints and index checks detect changed inputs.

Narrative chunk size is a word target · table rows preserve their labels and conditions · original source bytes remain unchanged

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Retrieve the rule. Generate an answer. Check the claims before release.

A small local pipeline makes evidence, control decisions and failure outcomes inspectable.

PREPARED OFFLINE Verified passages + keyword statistics + normalized vectors

UNDERSTAND	RETRIEVE	SELECT	GENERATE	CHECK	RESPOND
Question context + search scope	BM25 + MiniLM fusion + rerank	≤3 passages ~520 words	Qwen3-4B Q4 local CPU	Citations Numbers Support	Cited answer or source review

Weak evidence → clarify or report insufficient support.

Failed checks → one revision → withhold if still failing.

Why retrieval-augmented generation?

Replaceable source material and explicit citations.

Why hybrid search?

Exact identifiers plus paraphrases; jointly rerank.

Why this runtime?

Local CPU, no API key; small in-memory index.

MiniLM-L6 embeddings and reranking · DeBERTa NLI support screening · source identity and claim-local citations

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Context, citations and review in one conversational workflow.

A follow-up retains the user's situation while retrieving new supporting evidence.

Search and scope

Procedure library and guided starters; selected documents restrict retrieval.

Follow-up resolution

Retain the situation. A new topic changes scope; every turn retrieves evidence.

Inspect and review

Open citations, checks and activity; flag a response and export the investigation.

RECORDED CONVERSATION / PR63

What is the overnight storage limit for a coin-counting machine?

The overnight storage limit for a coin-counting machine is $5,250 [S1].

And is monthly testing done in single or dual control?

Monthly testing is done in single control. [S2]

S1: overnight storage → S2: monthly testing

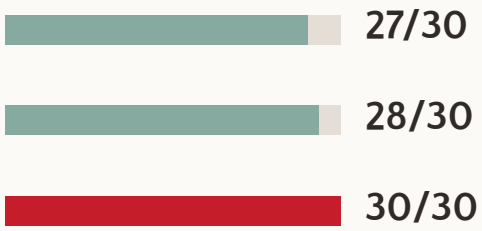
Every turn retrieves evidence · prior generated answers are not source material · scope, reset, session isolation, feedback and export are supported

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Evaluate retrieval and response quality as separate measures.

Measured local regression results; independent review is still required for an operational decision.

RELEVANT PASSAGE IN TOP FIVE



23/30 answers pass the expected-fact proxy

Phrase matching can miss valid wording or pass an incorrect answer.

WHAT TO MEASURE / WHAT A REVIEWER MUST JUDGE

Accuracy	Citations, numbers and support screens.	Is every instruction correct for this situation?
Relevance	Passage ranking and direct response.	Does it answer the exact question?
Completeness	23/30 expected-fact proxy.	Are material conditions and exceptions included?
Conciseness	28-word mean draft; bounded output.	Is it brief without losing necessary guidance?

Next: two independent reviewers, unseen scenarios, critical-error checks.

6/35 generated evaluations: source only

2/4 original follow-ups: fact-proxy pass

CPU p50 / p95: 13.8s / 23.9s

30 procedure questions + 4 two-turn conversations + 6 selected guardrails · AI-authored/source-checked labels · cold load excluded · draft length n=32

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Demonstrate answers, follow-ups and evidence review.

Show the functional chat with procedure questions, changed conditions and a clear review path.

DEMONSTRATION SEQUENCE

01 Ask a routine rule

Coin overnight limit → monthly-testing follow-up.

$5,250 overnight; single control for testing. [PR63]

02 Change a condition

Emailed notary document → requester is present.

The emailed-document restriction still applies. [PR1000]

03 Inspect an exception

Newer account → Private Bank Experience exception.

$1,000/day base rule; inspect the exception. [PR1949]

WHAT TO OBSERVE

Grounding and traceability

Inspect the exact passage and the checks behind the released response.

Conversation continuity

Preserve applicable conditions; change topics without carrying old evidence.

Bounded failure handling

Clarify missing context or show sources when an answer cannot pass checks.

Next: independent quality review, broader exception tests and deployment controls.

Generative AI: development assistance and local runtime answering · recorded outputs and test results are checked against source procedures